

ADVANCED COGNITIVE NEUROSCIENCE

INTRODUCTION TO CODING PIPELINE

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PLAN FOR TODAY

1. Presentation and discussion of Van Vliet (2020) and coding practices during the course
2. Live-coding to introduce core principles
3. Hands-on coding!

Van Vliet, M. (2020). Seven quick tips for analysis scripts in neuroimaging. *PLOS Computational Biology*, 16(3), e1007358. <https://doi.org/10.1371/journal.pcbi.1007358>



AIMS

The main goal of this class today, is to introduce you to the “bookkeeping” aspect of neuroimaging analysis.



BOOKKEEPING

Today

Parameters

Naming conventions

Structuring scripts

Quality control

Automatisation

Documentation

File organisation

Reproducibility

CONTENT

Filtering

Statistics

ICA

Source estimation

Evoked responses

Multivariate analysis

Beamforming

Artefact rejection

Comparing analysis choices

Minimum norm estimation



WHY IS GOOD BOOKKEEPING NECESSARY?

Many analysis steps, some computationally heavy, which rely on previous steps....

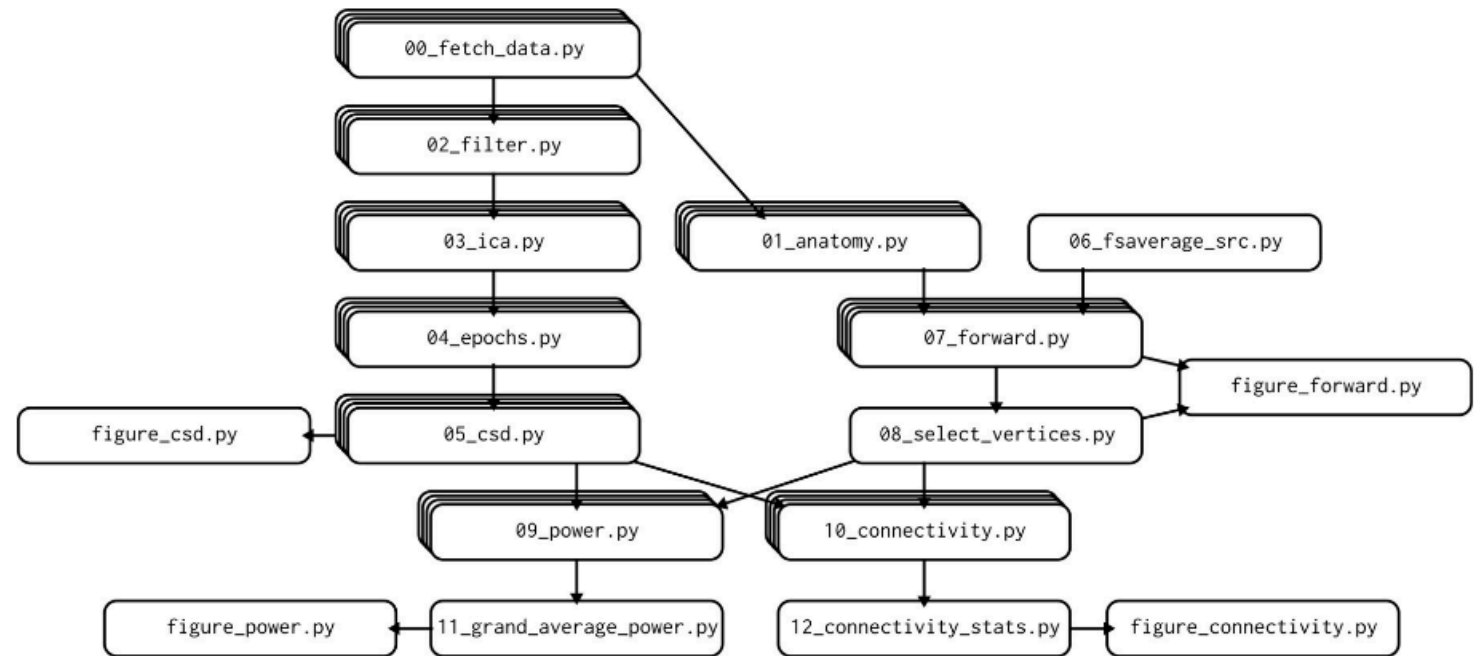


Fig 2. Dependency graph showing how the output of one script is used by another. Stacked boxes indicate scripts that are run for each participant.

TIPS FROM VAN VLIET (2020)

Tip 1: Each analysis step is one script

Tip 2: A script either processes a single recording or aggregates across recordings, never both

Tip 3: One master script to run the entire analysis

Tip 4: Save all intermediate results

Tip 5: Visualize all intermediate results

Tip 6: Each parameter and file name is defined only once

Tip 7: Distinguish files that are part of the official pipeline from other scripts



CLASS DISCUSSION



In what scenarios would these tips save time or prevent problems?

Tip 1: Each analysis step is one script

Tip 2: A script either processes a single recording or aggregates across recordings, never both

Tip 3: One master script to run the entire analysis

Tip 4: Save all intermediate results

Tip 5: Visualize all intermediate results

Tip 6: Each parameter and file name is defined only once

Tip 7: Distinguish files that are part of the official pipeline from other scripts



I WANT YOUR FEEDBACK

ALL THE TIME!

What I want you to get out of these classes

- Meet the course requirements → analyse the data for your portfolio report
- Build transferable skills → write clean, reproducible analysis scripts
- Explore the data → experiment and develop an intuition for the data and the effect of different analysis choices

Trade-off: Reproducible workflows (and also playing around with the data) involve quite a few files and things to keep track of!

If something isn't working we can adjust:

- More/less live coding
- More/less independent experimentation
- More/less focus on bookkeeping vs. content



LIVE CODING

1. Cloning GitHub repository
2. Setting up virtual environment
3. FileNames (+ naming conventions)
4. MNE reports
5. Create folder for own scripts!



HANDS ON CODING

Taking inspiration from the notebooks and skeleton scripts, start working on setting up the scripts

- 03_filter.py
- 04_ica.py
- 05_epochs.py

Remember to also create the `params.py` and copy needed utils!

... to be continued next class!

